

1. Executive Summary

		Traffic Summary		Count	
Metric	Value	Total Flows Observed		16016	
Total Threats Mitigated	16	Total Packets Analyzed		8153673666	
High Priority	11	Malicious Packets		8128770058	
Low Priority	5	True Negatives Passed		11287	
Manual Releases	0				
Manual Blocks	0	Attack Vector		Count	
		ICMP Flood		0	
ML Metrics	Value	SYN Flood		0	
False Positives	0	UDP Flood		13	
FP Rate	0.00%	Uncertain		3	
IF Precision	100.00%				
IF Recall	95.47%	Action		Count	
IF F1-Score	97.68%	Quarantined		3	
IF Accuracy	98.61%	Time Ban		3	
		Blackhole		0	
		Released		2	

2. Performance Benchmark

2a. Isolation Forest: Anomaly Detection

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.47%	Share of actual attacks successfully flagged
F1-Score	97.68%	Balanced measure combining Precision and Recall
Accuracy	98.61%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.53%	Actual attacks that went undetected
True Positive Rate (TPR)	95.47%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4700	FN 223
Actual: Normal	FP 0	TN 11064

2b. Random Forest: Attack Classification

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	100.00%	Share of actual attacks successfully flagged
F1-Score	100.00%	Balanced measure combining Precision and Recall
Accuracy	100.00%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	0.00%	Actual attacks that went undetected
True Positive Rate (TPR)	100.00%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: SYN	Predicted: ICMP	Predicted: UDP
Act: SYN	0	0	0
Act: ICMP	0	0	0
Act: UDP	0	0	4700

2c. Hybrid Cascaded

Metric	Value	Description
Precision	100.00%	Share of flagged anomalies that were genuine attacks
Recall (TPR)	95.47%	Share of actual attacks successfully flagged
F1-Score	97.68%	Balanced measure combining Precision and Recall
Accuracy	98.61%	Overall proportion of correct anomaly decisions
False Positive Rate (FPR)	0.00%	Normal traffic incorrectly flagged as an attack
False Negative Rate (FNR)	4.53%	Actual attacks that went undetected
True Positive Rate (TPR)	95.47%	Same measure as Recall, attacks correctly flagged
True Negative Rate (TNR)	100.00%	Normal traffic correctly identified as safe

	Predicted: Attack	Predicted: Normal
Actual: Attack	TP 4700	FN 223
Actual: Normal	FP 0	TN 11064

2d. Response Latency

Metric	Value	Description
Detection Time	715.8 ms	Average time taken to detect an attack
Mitigation Response Time	5.1 ms	Interval between the anomaly flag and the FlowMod blocking rule install

2e. Controller Resource Overhead

Metric	Value	Description
Controller CPU (Baseline)	1.02%	Ryu controller CPU usage under normal traffic
Controller CPU (Active Attack)	7.24%	Ryu controller average CPU usage during a DDoS simulation
Controller CPU (Mitigation)	7.00%	Ryu controller CPU usage during simultaneous detection and mitigation

3. Offences Summary

Source IP	Sessions	Max Offences	First Seen	Last Seen
10.0.0.21	5	4	2026-09-26 18:51:15	2026-09-26 18:53:55
10.0.0.22	5	4	2026-09-26 18:51:09	2026-09-26 18:53:50
10.0.0.20	4	3	2026-09-26 18:51:10	2026-09-26 18:53:51

4. Chronological Mitigation Log

Timestamp	Source IP	Class	Vector	Confidence	Priority	Action
2026-09-26 18:51:09	10.0.0.22	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:51:10	10.0.0.20	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:51:15	10.0.0.21	DDoS	Uncertain	-	Low	Quarantined
2026-09-26 18:51:20	10.0.0.22	DDoS	UDP Flood	76.7%	High	Time Ban (30s)
2026-09-26 18:51:20	10.0.0.22	DDoS	UDP Flood	76.7%	High	Time Ban
2026-09-26 18:51:21	10.0.0.20	DDoS	UDP Flood	79.5%	High	Time Ban (30s)
2026-09-26 18:51:21	10.0.0.20	DDoS	UDP Flood	79.5%	High	Time Ban
2026-09-26 18:51:25	10.0.0.21	DDoS	UDP Flood	78.7%	High	Time Ban (30s)
2026-09-26 18:51:26	10.0.0.21	DDoS	UDP Flood	78.7%	High	Time Ban
2026-09-26 18:51:50	10.0.0.22	DDoS	UDP Flood	76.7%	High	Released
2026-09-26 18:51:55	10.0.0.21	DDoS	UDP Flood	84.6%	High	Released
2026-09-26 18:52:00	10.0.0.22	DDoS	UDP Flood	79.3%	High	Time Ban (1m)
2026-09-26 18:52:06	10.0.0.21	DDoS	UDP Flood	85.8%	Critical	Time Ban (1m)
2026-09-26 18:52:21	10.0.0.20	DDoS	UDP Flood	79.5%	High	Time Ban (1m)
2026-09-26 18:53:07	10.0.0.21	DDoS	UDP Flood	85.8%	High	Time Ban (2m)
2026-09-26 18:53:50	10.0.0.22	DDoS	UDP Flood	76.0%	Critical	Time Ban (2m)

5. IP Attack History (Completed Sessions)

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.21	UDP Flood	0.6265	86.2%	Critical	Phase 2	1	10s	2026-09-26 18:53:55	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6448	77.9%	Critical	Phase 2	1	11s	2026-09-26 18:53:51	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6398	76.0%	Critical	Phase 2	1	11s	2026-09-26 18:53:50	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6264	85.8%	High	Phase 2	4	1m 12s	2026-09-26 18:53:07	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6083	81.1%	High	Phase 2	4	1m 10s	2026-09-26 18:53:00	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6308	79.5%	High	Phase 2	3	1m 11s	2026-09-26 18:52:21	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6357	85.8%	Critical	Phase 2	3	11s	2026-09-26 18:52:06	Attack persisted - escalating to time ban

Source IP	Vector	IF Score	Conf	Priority	Phase	Offences	Duration	Unblocked At	Reason
10.0.0.22	UDP Flood	0.6371	79.3%	High	Phase 2	3	10s	2026-09-26 18:52:00	Attack persisted - escalating to time ban
10.0.0.21	UDP Flood	0.6451	84.6%	High	Phase 2	1	40s	2026-09-26 18:51:55	Ban Expired
10.0.0.20	UDP Flood	0.6321	79.5%	High	Phase 2	2	41s	2026-09-26 18:51:51	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.5803	76.7%	High	Phase 2	1	41s	2026-09-26 18:51:50	Ban Expired
10.0.0.21	UDP Flood	0.6458	78.7%	High	Phase 2	1	10s	2026-09-26 18:51:25	Attack persisted - escalating to time ban
10.0.0.20	UDP Flood	0.6317	79.5%	High	Phase 2	1	11s	2026-09-26 18:51:21	Attack persisted - escalating to time ban
10.0.0.22	UDP Flood	0.6402	76.7%	High	Phase 2	1	11s	2026-09-26 18:51:20	Attack persisted - escalating to time ban

Verification and Approval

I hereby verify that the findings in this incident report are accurate and that the listed mitigation actions were carried out under my supervision.

SIGNATURE OVER PRINTED NAME

Network Administrator

DATE